ENERGIA: USB-C, carga de bateria, reparto de carga y regulador 3.3 V

J1 — USB-C 16P
VBUS (A4, B9), CC1 (A5), CC2 (B5), D- (A7), D- (B7), D+ (A6), D+ (B6), SBU1 (A8), SBU2 (B8), SHIELD, GND (A1, B12, S1)

U1 — TP4056
VBUS, CE, CHRG_N, STDBY_N, Vcc, BAT, GND, TEMP, PROG, EPAD
BAT → VBAT; PROG → R3 2k → GND

J2 — BATERIA LiPo
VBAT, GND

Usar bateria con circuito de proteccion integrado

Q1 — AO3401A
VBUS, VBAT, VSYS

U2 — AP2112K-3.3
VSYS, PWR_EN, VIN, EN, VOUT, GND → 3V3_REG

D3 — B5819W
VSYS, VBUS

SW1 — ENCENDIDO
PWR_EN, VSYS, GND

JP1 — 3V3
3V3_REG → +3V3

R1 5.1k CC1 → GND
R2 5.1k CC2 → GND
C1 10uF VBUS → GND
R3 2k PROG → GND
C2 10uF VBAT → GND
R6 100k VBUS → GND
C3 1uF VSYS → GND
C4 10uF 3V3_REG → GND
R15 100k PWR_EN → GND
R11 470k VBAT → BAT_SENSE
R12 470k BAT_SENSE → GND
C11 100nF BAT_SENSE → GND
R4 1k VBUS → LED_CHG_A
R5 1k VBUS → LED_FULL_A

D1 — CARGANDO (rojo)
CHRG_N, LED_CHG_A

D2 — CARGA COMPLETA (verde)
STDBY_N, LED_FULL_A

Puntos de prueba y banderas de alimentacion

TP1 VBUS, TP2 VBAT, TP3 VSYS, TP4 +3V3, TP5 GND, TP6 EN, TP7 BTN_MENU, TP8 UART_TX, TP9 UART_RX

PWR_FLAG VBUS, PWR_FLAG VSYS, PWR_FLAG +3V3, PWR_FLAG GND

PANTALLA (conector PROVISIONAL)

J3 — LCD 2.4in ST7789V
GND, +3V3, LCD_SCK, LCD_MOSI, LCD_CS, LCD_DC, LCD_RST, LCD_LEDA, LCD_LEDK

Q2 — AO3400A
LCD_BL, LCD_LEDK, GND

R13 10R (por definir) +3V3 → LCD_LEDA
R14 100k LCD_BL → GND

Pines y resistencia de luz de fondo:
se definen al tener el panel en la mano

microSD (SPI)

J4 — microSD
SD_CS, SD_MOSI, +3V3, SD_SCK, GND, SD_MISO, DAT2, DAT3/CD, CMD, VDD, CLK, VSS, DAT0, DAT1, SHIELD, GND

R8 10k +3V3 → SD_CS
R9 10k +3V3 → SD_MISO
C8 100nF +3V3 → GND
C12 10uF +3V3 → GND

ESP32-S3-WROOM-1-N16R8

U3 — ESP32-S3-WROOM-1-N16R8
+3V3, EN, 3V3, GND, PSRAM
BTN_MENU 27, BAT_SENSE 28, BTN_L 38, BTN_UP 1, BTN_DOWN 5, BTN_LEFT 6, BTN_RIGHT 7, BTN_A 12, LCD_DC 17, LCD_CS 18, LCD_MOSI 19, LCD_SCK 20, LCD_BL 21, LCD_RST 22, SD_MOSI 8, SD_SCK 9
TXD0 37 UART_TX, RXD0 36 UART_RX, IO17 10 SD_MISO, IO18 11 SD_CS, USB_D- 13 USB_DM, USB_D+ 14 USB_DP, IO21 23 BTN_B, IO38 31 BTN_R, IO39 32 BTN_OPTION, IO40 33 I2S_BCLK, IO41 34 I2S_LRCLK, IO42 35 I2S_DIN, IO45 26, IO46 16, IO47 24 BTN_START, IO48 25 BTN_SELECT

SW2 — RESET
EN → GND

R7 10k +3V3 → EN
C5 1uF EN → GND
C6 10uF +3V3 → GND
C7 100nF +3V3 → GND

BOTONES (cada uno de su GPIO a GND, pull-up interno)

SW3 BTN_UP → GND — UP
SW4 BTN_DOWN → GND — DOWN
SW5 BTN_LEFT → GND — LEFT
SW6 BTN_RIGHT → GND — RIGHT
SW7 BTN_A → GND — A
SW8 BTN_B → GND — B
SW9 BTN_START → GND — START
SW10 BTN_SELECT → GND — SELECT
SW11 BTN_MENU → GND — MENU
SW12 BTN_OPTION → GND — OPTION
SW13 BTN_L → GND — L
SW14 BTN_R → GND — R

MENU (GPIO 0): presionado al encender entra en modo de grabacion

AUDIO (I2S)

U4 — MAX98357A
VSYS, VDD, I2S_DIN, I2S_BCLK, I2S_LRCLK, BIN, BCLK, LRCLK, OUTP, OUTN, SPK_P, SPK_N, SD_MODE, AMP_SD, GAIN_SLOT, GND, EP_PAD

J5 — BOCINA BR
SPK_P, SPK_N

R10 1M VSYS → AMP_SD
C9 10uF VSYS → GND
C10 100nF VSYS → GND

GAIN sin conectar = 9 dB

Borrador generado. Sin probar en hardware.

Sheet: /
File: magsafe-console.kicad_sch

Title: MagSafe Console

Size: A2 | Date: | **Rev: 0.1**
KiCad E.D.A. 9.0.9 | Id: 1/1